

Body Performance Prediction - Project Report

Executive Summary

This project develops an intelligent machine learning system to analyze body performance using demographic, body composition, vital signs, and fitness measurements. It combines Exploratory Data Analysis (EDA), Classification, and Regression techniques.

Problem Statement

The project predicts the fitness class (A, B, C, D) of an individual and estimates broad jump distance based on physical characteristics and fitness indicators.

Dataset

13,393 records with 12 features including age, gender, height, weight, body fat percentage, blood pressure, grip strength, sit-and-bend, sit-ups count, broad jump distance, and fitness class.

Data Preparation

Validated data quality, checked missing values, removed duplicates, reviewed outliers, encoded categorical variables, and scaled numerical features before training.

Exploratory Data Analysis

Analyzed feature distributions, class balance, correlations, and relationships between body composition, fitness metrics, and body performance classes.

Machine Learning Models

Classification: MLP Classifier, Support Vector Machine (RBF), Decision Tree.

Regression: MLP Regressor and KNN Regressor.

Model Evaluation

Models were evaluated using multiple train/test splits (80/20, 70/30, 50/50) and K-Fold Cross Validation. Classification metrics included Accuracy, Precision, Recall, and F1-score. Regression metrics included MAE, RMSE, and R^2 .

Key Findings

- Broad Jump was the strongest predictor of body performance.
- Grip Strength and Sit-ups Count were highly influential features.
- Lower body fat was associated with higher fitness classes.
- Neural Networks and SVM achieved the best classification performance.
- Decision Trees balanced interpretability with competitive accuracy.
- Performance remained stable across different validation strategies.

Business Value

The workflow demonstrates a complete end-to-end machine learning pipeline, from data preparation to model evaluation, making it suitable as a portfolio project for Data Analysis and Machine Learning roles.

Future Improvements

Evaluate Random Forest, XGBoost, LightGBM, SHAP explainability, and deploy the best model using Streamlit or Flask.